VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Agile and Devops Lab Model exam

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Faculy Name: Balaganesh

School: SCORE

Slot: L3+L4

Payment processing system

Problem statement:

Ensure payments are processed securely and correctly

Tasks:

!.Develop payment validation logic

2.write Junit tests for transaction success/failure

3.implement CI/CD pipeline with automated testing

4.Deploy services on Docker and Kubernetes

Simple java application folder

Name	Status	Date modified	Type	Size
src		29-03-2026 10:09	File folder	
deployment		13-04-2026 10:10	Yaml Source File	1 KB
Dockerfile		13-04-2026 10:10	File	1 KB
pom		13-04-2026 10:10	Microsoft Edge HT...	2 KB

App.java

```
public class App {
   // Method to validate payment
   public static boolean processPayment(double amount, String cardNumber,
String cvv) {
      if (amount <= 0) {
         return false;
      }
      if (cardNumber == null || cardNumber.length() != 16) {
         return false;
      if (!cardNumber.matches("\\d+")) {
         return false;
      }
      if (cvv == null || cvv.length() != 3) {
         return false;
      }
      if (!cvv.matches("\\d+")) {
         return false;
      }
      return true;
   }
```

```
    public static void main(String[] args) {
        boolean result = processPayment(1000, "1234567812345678", "123");

        if (result) {
            System.out.println("Payment Successful");
        } else {
            System.out.println("Payment Failed");
        }
    }
}
```

AppTest.java

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
public class AppTest {
    @Test
    void testValidPayment() {
        assertTrue(App.processPayment(500, "1234567812345678", "123"));
    }
    @Test
    void testInvalidAmount() {
        assertFalse(App.processPayment(0, "1234567812345678", "123"));
    }
    @Test
    void testInvalidCardNumberLength() {
        assertFalse(App.processPayment(500, "1234", "123"));
```

```
    }
    @Test
    void testInvalidCardNumberCharacters() {
        assertFalse(App.processPayment(500, "1234abcd5678efgh", "123"));
    }
    @Test
    void testInvalidCVVLength() {
        assertFalse(App.processPayment(500, "1234567812345678", "12"));
    }
    @Test
    void testInvalidCVVCharacters() {
        assertFalse(App.processPayment(500, "1234567812345678", "abc"));
    }
}
```

Pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.demo</groupId>
 <artifactId>result-analytics</artifactId>
 <version>1.0</version>
 <properties>
  <maven.compiler.source>17</maven.compiler.source>
  <maven.compiler.target>17</maven.compiler.target>
```

```
 <junit.version>5.10.1</junit.version>
</properties>
<dependencies>
 <dependency>
  <groupId>org.junit.jupiter</groupId>
  <artifactId>junit-jupiter</artifactId>
  <version>${junit.version}</version>
  <scope>test</scope>
 </dependency>
</dependencies>
<build>
 <plugins>
  <plugin>
   <groupId>org.apache.maven.plugins</groupId>
   <artifactId>maven-compiler-plugin</artifactId>
   <version>3.11.0</version>
  </plugin>
  <plugin>
   <groupId>org.apache.maven.plugins</groupId>
   <artifactId>maven-surefire-plugin</artifactId>
   <version>3.2.5</version>
   <configuration>
    <useModulePath>false</useModulePath>
   </configuration>
  </plugin>
```

```
  <plugin>
   <groupId>org.apache.maven.plugins</groupId>
   <artifactId>maven-jar-plugin</artifactId>
   <version>3.4.1</version>
   <configuration>
    <archive>
     <manifest>
      <mainClass>App</mainClass>
     </manifest>
    </archive>
   </configuration>
  </plugin>
 </plugins>
</build>
</project>
```

Mvn clean test

```
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] ---------------------< com.demo:result-analytics >----------------------
[INFO] Building result-analytics 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ result-analytics ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ result-analytics ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vinay kumar reddy\OneDrive\??????\results-analytics\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ result-analytics ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ result-analytics ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vinay kumar reddy\OneDrive\??????\results-analytics\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ result-analytics ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ result-analytics ---
[INFO] No tests to run.
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  1.622 s
[INFO] Finished at: 2026-04-13T10:20:27+05:30
[INFO] ------------------------------------------------------------------------
```

Mvn clean package

```
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> mvn clean package
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.demo:result-analytics >----------------------
[INFO] Building result-analytics 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ result-analytics ---
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ result-analytics ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vinay kumar reddy\OneDrive\??????\results-analytics\src\main\resources
[INFO]
[INFO] --- compiler:3.11.0:compile (default-compile) @ result-analytics ---
[INFO] No sources to compile
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ result-analytics ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\vinay kumar reddy\OneDrive\??????\results-analytics\src\test\resources
[INFO]
[INFO] --- compiler:3.11.0:testCompile (default-testCompile) @ result-analytics ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:3.2.5:test (default-test) @ result-analytics ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ result-analytics ---
[WARNING] JAR will be empty - no content was marked for inclusion!
[INFO] Building jar: C:\Users\vinay kumar reddy\OneDrive\??????\results-analytics\target\result-analytics-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.006 s
[INFO] Finished at: 2026-04-13T10:20:41+05:30
[INFO] ------------------------------------------------------------------------
```

Target file

OneDrive > vinaykumar - Personal > เอกสาร > results-analytics >

Sort View ...

Name	Status	Date modified	Type	Size
src		29-03-2026 10:09	File folder	
target		13-04-2026 10:21	File folder	
deployment		13-04-2026 10:10	Yaml Source File	1 KB
Dockerfile		13-04-2026 10:10	File	1 KB
pom		13-04-2026 10:10	Microsoft Edge HT...	2 KB

OneDrive > vinaykumar - Personal > เอกสาร > results-analytics > target >

Sort View ...

Name	Status	Date modified	Type	Size
maven-archiver		13-04-2026 10:21	File folder	
result-analytics-1.0		13-04-2026 10:20	Executable Jar File	2 KB

DockerFile

```
FROM eclipse-temurin:17-jdk-alpine
WORKDIR /app
COPY target/*.jar app.jar
EXPOSE 8080
ENTRYPOINT ["java","-jar","app.jar"]
```

deployment.yaml

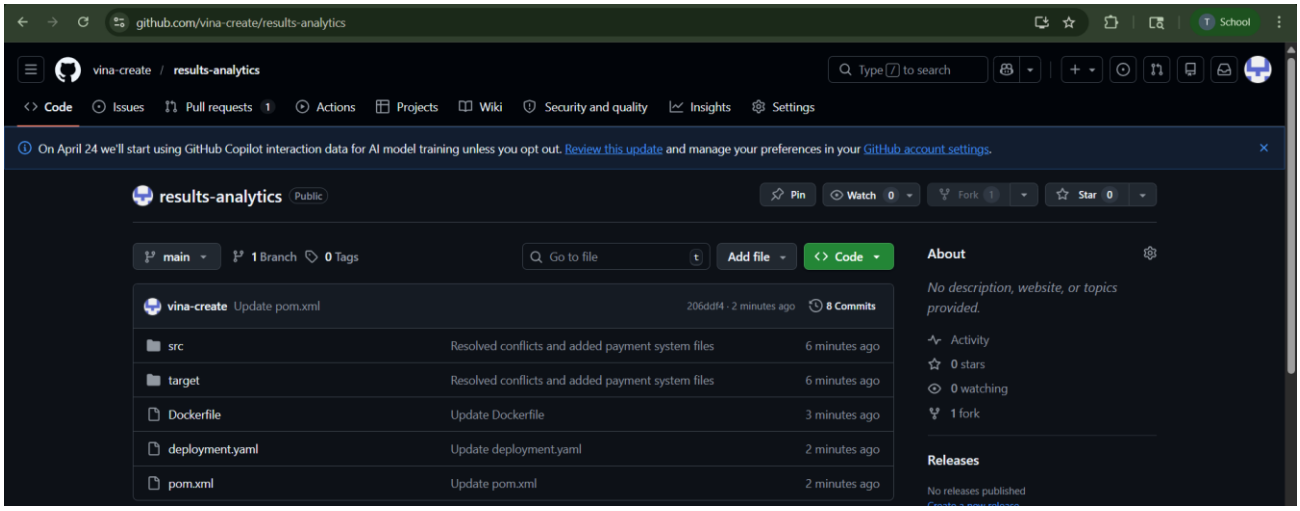
```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: java-test-app-deployment
  labels:
    app: java-test-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: java-test-app
  template:
    metadata:
      labels:
        app: java-test-app
    spec:
      containers:
        - name: java-test-app
          image: vinay3112/java-test-app:latest
```

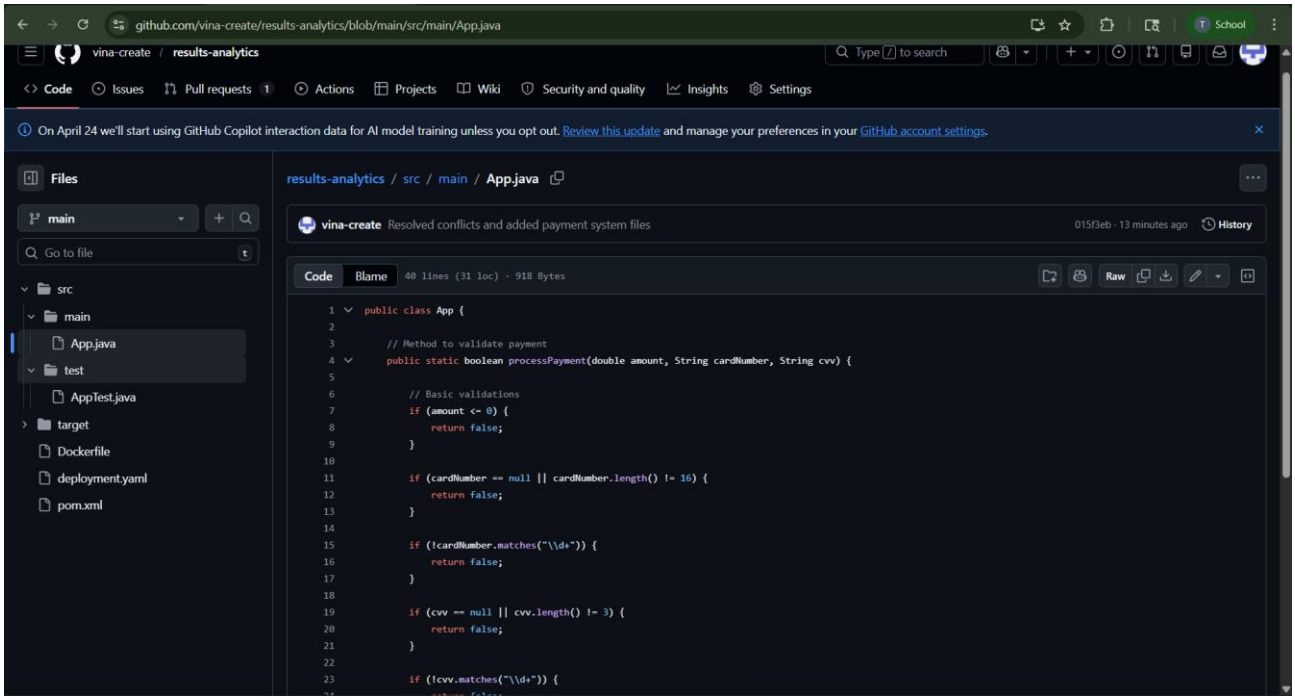
```
        imagePullPolicy: Always
        ports:
          - containerPort: 8080
---
apiVersion: v1
kind: Service
metadata:
  name: java-test-app-service
spec:
  type: NodePort
  selector:
    app: java-test-app
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30007
```

Push Code to GitHub

```
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> git add .
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> git commit -m "Resolved conflicts and added payment system files"
[detached HEAD 015f3eb] Resolved conflicts and added payment system files
 3 files changed, 45 insertions(+), 32 deletions(-)
```

```
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> git push origin main
Everything up-to-date
```





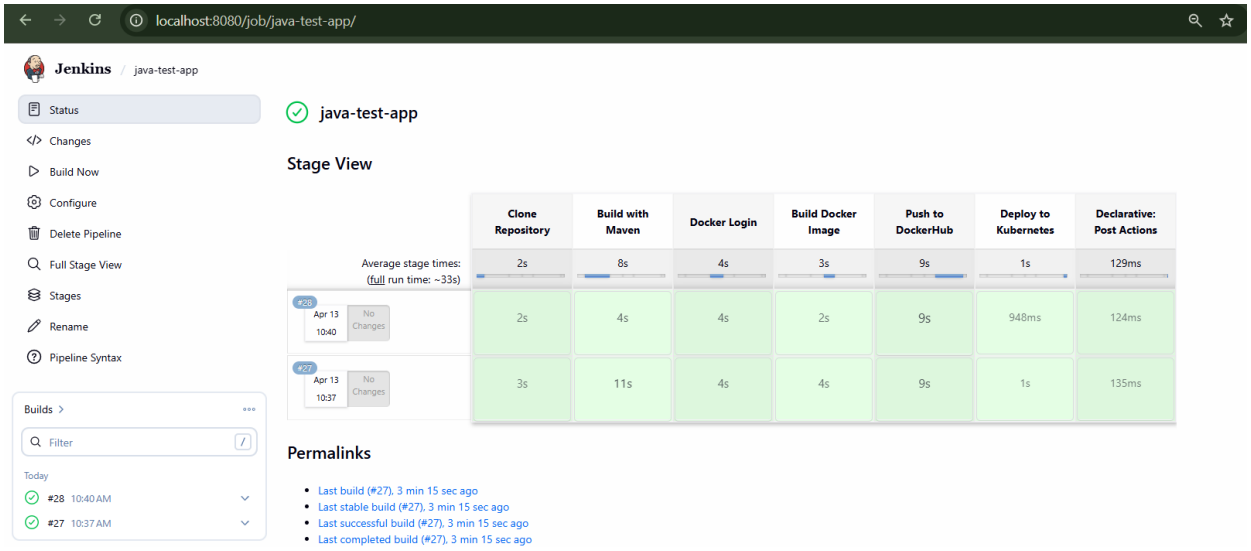
Jenkins pipeline job

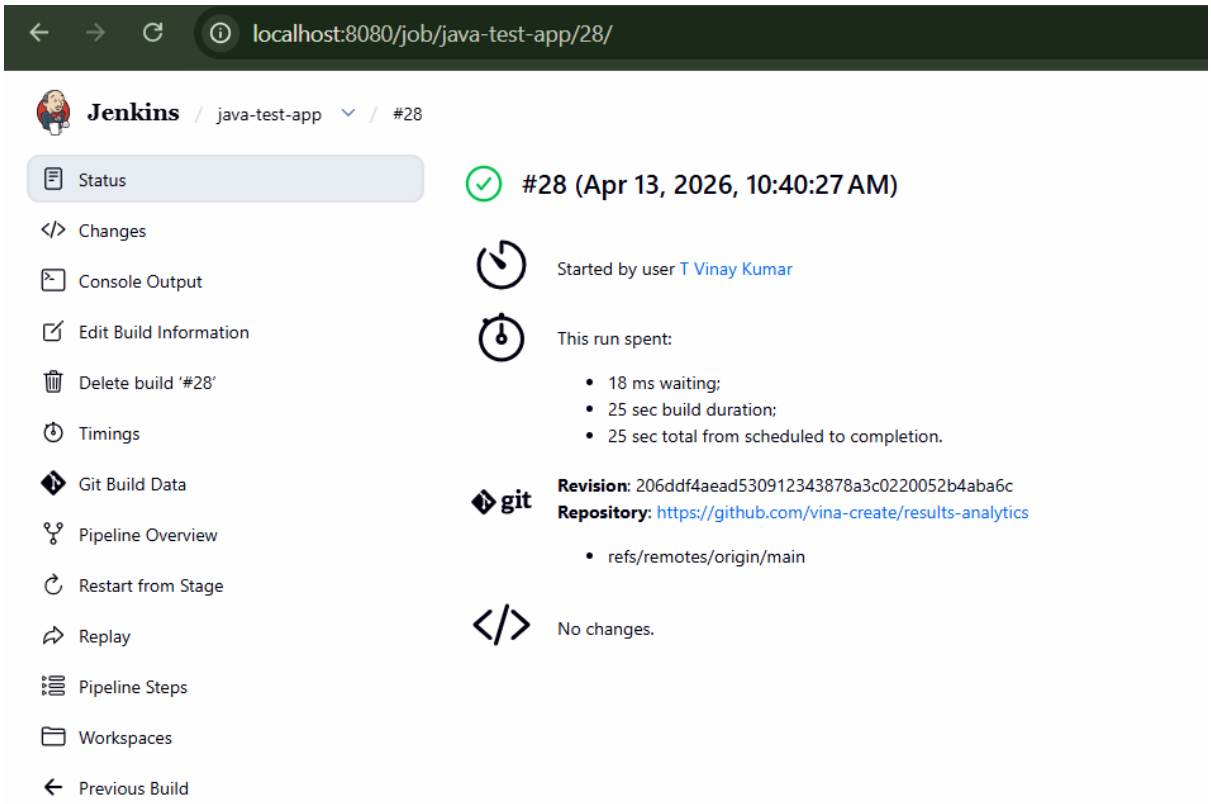
```
pipeline {
agent any
environment {
   DOCKER_IMAGE = "vinay3112/java-test-app:latest"
}
stages {
   stage('Clone Repository') {
      steps {
```

```
        git url: 'https://github.com/vina-create/results-analytics', branch: 'main'
    }
}
stage('Build with Maven') {
    steps {
        bat 'mvn clean package'
    }
}
// Direct Docker Login (bypasses Jenkins credential issues)
stage('Docker Login') {
    steps {
        bat '''
        docker logout
        docker login -u vinay3112 -p Docker008
        '''
    }
}
stage('Build Docker Image') {
    steps {
        bat "docker build -t vinay3112/java-app:latest ."
    }
}
stage('Push to DockerHub') {
    steps {
        bat "docker push vinay3112/java-app:latest"
    }
```

```
    }
   stage('Deploy to Kubernetes') {
    steps {
        bat '''
        set KUBECONFIG=C:\\Users\\vinay kumar reddy\\.kube\\config
        kubectl get nodes
        kubectl apply -f deployment.yaml
        '''
    }
}
}
post {
    success {
        echo 'CI/CD Pipeline executed successfully!'
    }
    failure {
        echo 'Pipeline failed. Please check logs.'
    }
}
}
```

Pipeline stage view plugin





Jenkins / java-test-app / #28

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#28'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces
- Previous Build

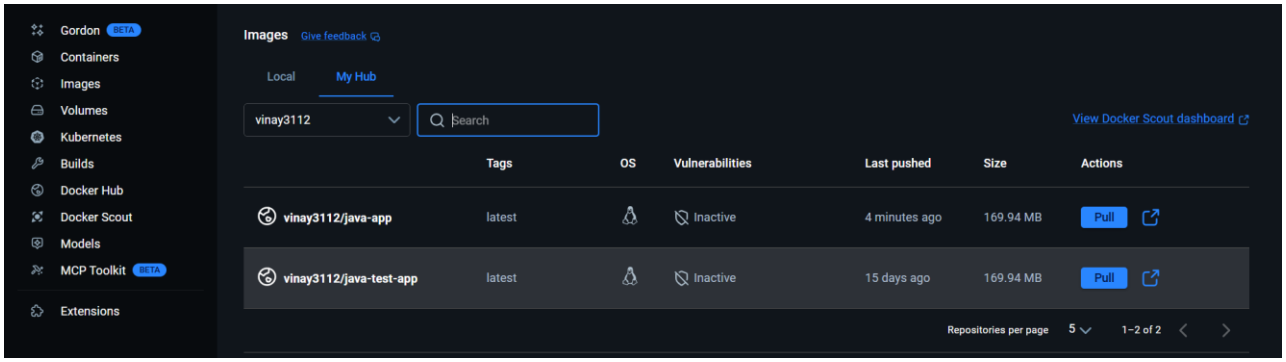
Console

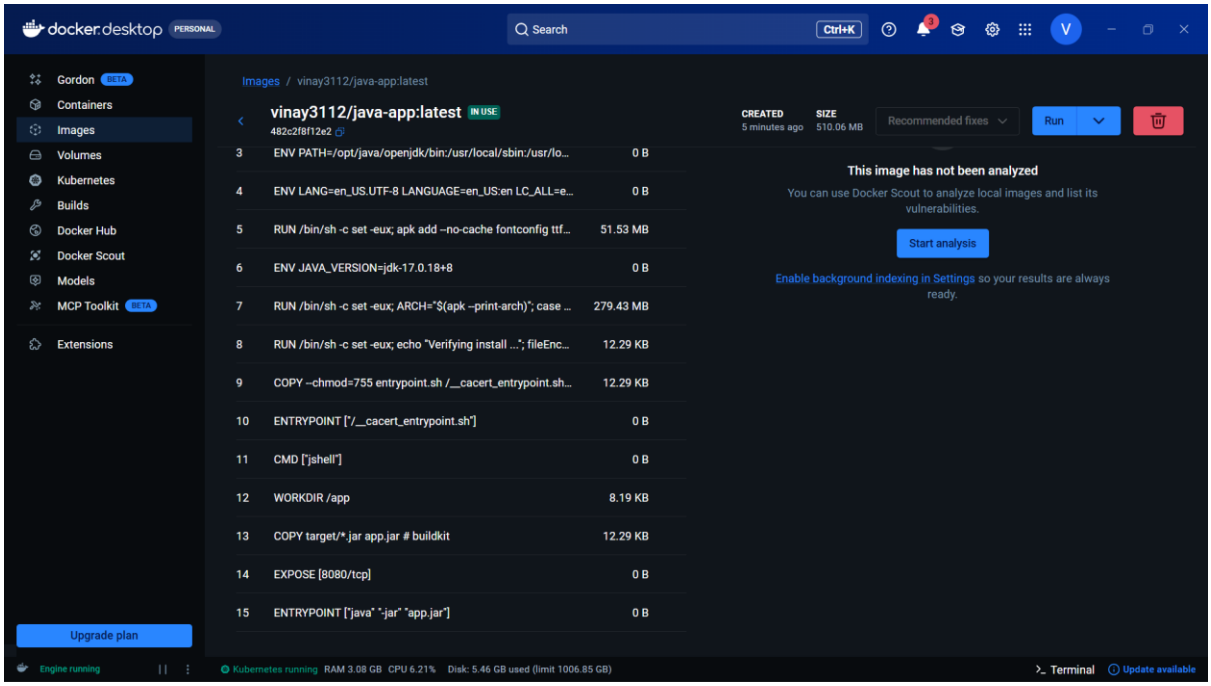
Download

```
Started by user T Vinay Kumar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\java-test-app
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Repository)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\cmd\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\java-test-app\.git # timeout=10
Fetching changes from the remote Git repository
 > C:\Program Files\Git\cmd\git.exe config remote.origin.url https://github.com/vina-create/results-analytics # timeout=10
Fetching upstream changes from https://github.com/vina-create/results-analytics
 > C:\Program Files\Git\cmd\git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > C:\Program Files\Git\cmd\git.exe fetch --tags --force --progress -- https://github.com/vina-create/results-analytics +refs/heads/*:refs/remotes/origin/* # timeout=10
 > C:\Program Files\Git\cmd\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 206ddf4aead530912343878a3c0220052b4aba6c (refs/remotes/origin/main)
 > C:\Program Files\Git\cmd\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -f 206ddf4aead530912343878a3c0220052b4aba6c # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -a -v --no-abbrev # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -D main # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -b main 206ddf4aead530912343878a3c0220052b4aba6c # timeout=10
Commit message: "Update pom.xml"
 > C:\Program Files\Git\cmd\git.exe rev-list --no-walk 206ddf4aead530912343878a3c0220052b4aba6c # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Build with Maven)
[Pipeline] bat
```

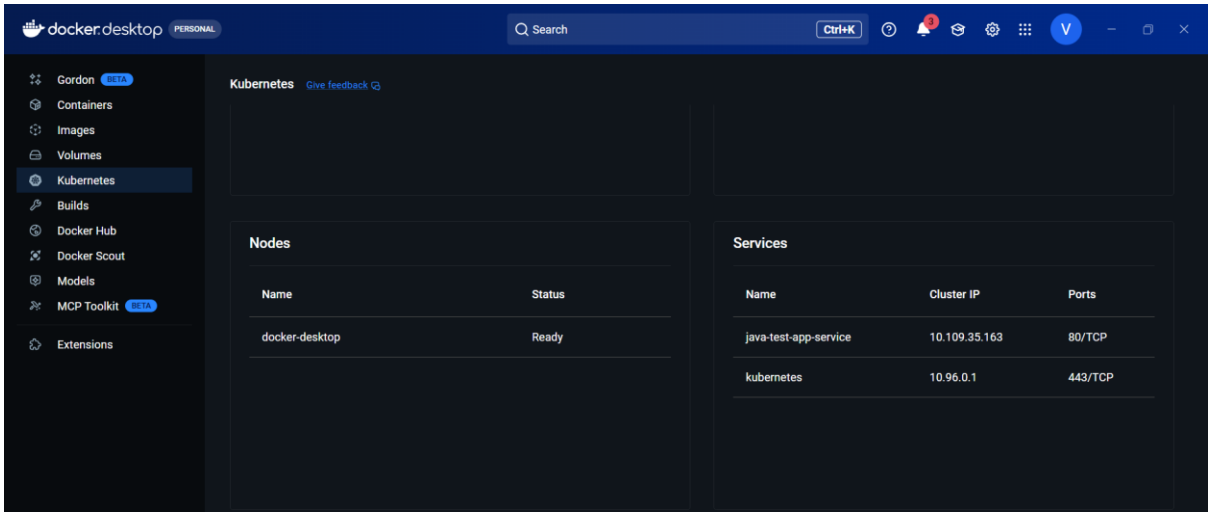
```
589002ba0eae: Layer already exists
88726997f454: Waiting
714ee9e66277: Waiting
88726997f454: Waiting
714ee9e66277: Waiting
88726997f454: Pushed
714ee9e66277: Pushed
latest: digest: sha256:482c2f8f12e22c7baaf60301593a03ab954cc2de8886fdbe8668f87a76ea9212 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\java-test-app>set KUBECONFIG=C:\Users\vinay kumar reddy\.kube\config

C:\ProgramData\Jenkins\.jenkins\workspace\java-test-app>kubectl get nodes
NAME             STATUS   ROLES           AGE   VERSION
docker-desktop   Ready    control-plane   14d   v1.34.1

C:\ProgramData\Jenkins\.jenkins\workspace\java-test-app>kubectl apply -f deployment.yaml
deployment.apps/java-test-app-deployment unchanged
service/java-test-app-service unchanged
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
CI/CD Pipeline executed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Docker Images





Kubernetes



Command Kubectl get svc

```
PS C:\Users\vinay kumar reddy\OneDrive\เอกสาร\results-analytics> kubectl get svc
NAME                    TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-test-app-service   NodePort    10.109.35.163   <none>        80:30007/TCP   14d
kubernetes              ClusterIP   10.96.0.1       <none>        443/TCP        14d
```